

**UNIT I**  
**BASICS OF C++**

**Course Objective - Cob1 :** To learn the basic concepts of OOP and getting started with c++.

**Course Outcome - CO1 :** Design a c++ programs with classes,objects and static member functions.

**Bloom's Taxonomy Level: K1, K2, K3, K4, K5, K6**

**Part A**

| <b>S.<br/>No.</b> | <b>Question</b>                                                                                                                                                                                                                                                                                                       | <b>BTL*</b> |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>1.</b>         | <b>Who developed C++ and What are the four main principles of Object-Oriented Programming?</b><br>C++ was developed by Bjarne Stroustrup in 1979 at Bell Labs.Encapsulation, Abstraction, Inheritance, and <b>Polymorphism</b>                                                                                        | <b>K1</b>   |
| <b>2.</b>         | <b>List out the characteristics of OOP.</b><br>1. Programs are divided into objects.<br>2. Data is hidden.<br>3. Objects communicate with each other, by sending messages and receiving responses.<br>4. It follows bottom-up approach.                                                                               | <b>K1</b>   |
| <b>3.</b>         | <b>List down the basic concepts of OOP.</b><br>1. Objects<br>2. Classes<br>3. Data abstraction and encapsulation<br>4. Inheritance<br>5. Polymorphism<br>6. Dynamic binding<br>7. Message passing                                                                                                                     | <b>K1</b>   |
| <b>4.</b>         | <b>Define an object.</b><br>Objects are the basic run time entities in an object oriented system.<br>They may represent a person, a place or any item that a program has to handle                                                                                                                                    | <b>K1</b>   |
| <b>5.</b>         | <b>Define Object Oriented Programming.</b><br>OOP is a method of implementation in which programs are organized as co-operative collection of objects, each of which represents an instance of some class and whose classes are all members of a hierarchy of classes united through the property called Inheritance. | <b>K1</b>   |
| <b>6.</b>         | <b>Define a class.</b><br>A class is a collection of objects with similar attributes and operations.<br>Eg. Class – Account<br>It will create an object savings_account belonging to the class Account                                                                                                                | <b>K1</b>   |

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|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 7.  | <b>Define Encapsulation.</b><br>The wrapping up of data and functions into a single unit is known as encapsulation. The data is kept safe from external interference and misuse.                                                                                                                                                                                                                       | K1 |
| 8.  | <b>Define Data hiding?</b><br>The isolation of data from direct access by the program is called as data hiding or information hiding.                                                                                                                                                                                                                                                                  | K1 |
| 9.  | <b>Define Abstraction and ADT.</b><br>It refers to the act of representing essential features without including the background details.<br>The classes which are using the concept of data abstraction is known as abstract data types (ADT).                                                                                                                                                          | K1 |
| 10. | <b>Define data member and member function?</b><br>The attributes which are holding the information is known as data members.<br>The functions that operate on data member are sometimes called as member function                                                                                                                                                                                      | K1 |
| 11. | <b>Define Inheritance?</b><br>Inheritance is the process by which objects of one class acquire the properties of objects of another class. The new derived class inherits the members of the base class and also adds its own.                                                                                                                                                                         | K1 |
| 12. | <b>Define Polymorphism?</b><br>It allows a single name/operator to be associated with different operations depending on the type of data passed to it. An operation may exhibit different behaviors in different instances.                                                                                                                                                                            | K1 |
| 13. | <b>What is message passing?</b><br>It is the process of invoking an operation on an object. In response to a message, the corresponding function is executed in the object.                                                                                                                                                                                                                            | K2 |
| 14. | <b>List down the benefits of OOP?</b><br>1. The principle of data hiding helps to build secure programs.<br>2. Through inheritance, redundant code is eliminated.<br>3. Software complexity can be easily managed.                                                                                                                                                                                     | K1 |
| 15. | <b>List down the applications of OOP.</b><br>1. Real time system<br>2. Simulation and modeling<br>3. AI and expert system<br>4. Neural network programming.<br>5. CAD/CAM systems.                                                                                                                                                                                                                     | K1 |
| 16. | <b>Define Array and Give the syntax of Array Initialization with an example.</b><br>Array is a collection of data of the same data type, stored at a contiguous memory location. Data-type array name[size]=(list of values separated by comma};<br>(eg.) int mark[5]={12,13,14,11,10};                                                                                                                | K1 |
| 17. | <b>Define string and Give four examples for String Manipulation functions.</b><br>Strings are used for storing text/characters. For example, "Hello World" is a string.<br>strlen( ) – finds the length of string<br>strcpy( ) – copies the contents of one string to another<br>strcat( ) – concatenates two strings into one single string<br>strupr( ) – converts a lower case string to upper case | K2 |

**Part B**

| <b>S.No.</b> | <b>Question</b>                                                                          | <b>BTL*</b> |
|--------------|------------------------------------------------------------------------------------------|-------------|
| <b>1.</b>    | Explain the key principles of Object-Oriented Programming (OOP) with examples in C++.    | K2          |
| <b>2.</b>    | Write a detailed note on constructors and destructors in C++ with suitable program       | K2          |
| <b>3.</b>    | Elaborate on parameterized constructor with an example.                                  | K6          |
| <b>4.</b>    | What is copy constructor and Give an example program                                     | K3          |
| <b>5.</b>    | What Static member and member function with an example program.                          | K2          |
| <b>6.</b>    | Discuss the fundamentals of arrays in C++ and provide an example of their usage.         | K3          |
| <b>7.</b>    | How are strings handled in C++? Discuss the operations that can be performed on strings. | K6          |
| <b>8.</b>    | Develop the simple string program by using string cat and concatenation                  | K6          |